

UNIVERSITY OF MADRAS

M.Sc. DEGREE PROGRAMME IN COMPUTER SCIENCE

SYLLABUS WITH EFFECT FROM 2023-2024

Title of the Paper	Advanced Computer Architecture		
Elective –I - Theory	I Year & I Semester	Credit:3	436E1C

Objectives:

- To introduce the students to the recent trends in the field of Computer Architecture and identify performance related parameters.
- To learn the different multiprocessor issues.
- To expose the different types of multicore architectures.
- To understand the design of the memory hierarchy.

Outcomes:

1.	Identify the limitations of Instruction-level parallelism (ILP).	K1, K2
2.	Discuss the issues related to multiprocessing and suggest solutions	K2, K3
3.	Point out the salient features of different multicore architectures and how they exploit parallelism.	K3, K4
4.	Discuss the various techniques used for optimizing the cache performance	K4, K5
5.	Design hierarchical memory system and Point out how data level parallelism is exploited in architectures	K5, K6
K1-Remember;K2-Understand;K3-Apply;K4-Analyze;K5-Evaluate; K6-Create		

UNIT I FUNDAMENTALS OF COMPUTER DESIGN AND ILP

Fundamentals of Computer Design – Measuring and Reporting Performance – Instruction Level Parallelism and its Exploitation – Concepts and Challenges –Exposing ILP - Advanced Branch Prediction - Dynamic Scheduling - Hardware-Based Speculation - Exploiting ILP - Instruction Delivery and Speculation - Limitations of ILP - Multithreading

UNIT II MEMORY HIERARCHY DESIGN

Introduction – Optimizations of Cache Performance – Memory Technology and Optimizations – Protection: Virtual Memory and Virtual Machines – Design of Memory Hierarchies – Case Studies.

UNIT III MULTIPROCESSOR ISSUES

Introduction- Centralized, Symmetric and Distributed Shared Memory Architectures – Cache Coherence Issues – Performance Issues – Synchronization – Models of Memory Consistency – Case Study-Interconnection Networks – Buses, Crossbar and Multi-stage Interconnection Networks

UNIT IV MULTICORE ARCHITECTURES

Homogeneous and Heterogeneous Multi-core Architectures – Intel Multicore Architectures – SUN CMP architecture – IBM Cell Architecture. Introduction to Warehouse-scale computers

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Architectures- Physical Infrastructure and Costs- Cloud Computing –Case Study- Google Warehouse-Scale Computer.

UNIT V VECTOR, SIMD AND GPU ARCHITECTURES

Introduction-Vector Architecture – SIMD Extensions for Multimedia – Graphics Processing Units – Case Studies – GPGPU Computing – Detecting and Enhancing Loop Level Parallelism-Case Studies.

Recommended Texts:

1. Darryl Gove ;Multicore Application Programming: For Windows, Linux, and Oracle Solaris, Pearson, 2011

Reference Books:

1. David B. Kirk, Wen-mei W. Hwu, – Programming Massively Parallel Processors, Morgan Kauffman, 2010
2. David E. Culler, Jaswinder Pal Singh, – Parallel computing architecture : A hardware/software approach , Morgan Kaufmann /Elsevier Publishers, 1999
3. John L. Hennessey and David A. Patterson, – Computer Architecture – A Quantitative Approach, Morgan Kaufmann / Elsevier, 5th edition, 2012.
4. Kai Hwang and Zhi.Wei Xu, – Scalable Parallel Computing, Tata McGraw Hill, NewDelhi, 2003

Web References:

1. <https://www.youtube.com/watch?v=v7iefsovo9M&list=PLwdnzlV3ogoWJhBxBYu-K4l-q-nNHd24D>

Mapping with Programme Outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10
CO1	M	S	M	S	S	M	L	M	L	S
CO2	S	M	S	L	M	L	M	L	M	S
CO3	S	S	L	M	S	S	L	M	L	M
CO4	M	L	S	M	M	L	S	L	M	S
CO5	S	S	M	S	L	M	L	M	S	L

S-Strong M-Medium L-Low